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CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No :2

TITLE : Develop aJava application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Code

```
class Student {
  String name;
  int age;

  Student(String n, int a) {
    name = n;
    age = a;
  }

  void display() {
    System.out.println("name : " + name + ", age : " + age);
  }
}

class Counter {
  static int count = 0;
```

```java
 Counter () {
 count++;
 System.out.println("Object : " + count);
 }
}

class Calculator {
 static int add(int a, int b) {
 return a + b;
 }
}

class Outer {
 class Inner {
 void display () {
 System.out.println("Inner Class Method");
 }
 }
}

public class Main {
 int id;
 String name;

 public static void main (String[] args) {
 // Defining class in java
 System.out.println("Defining class in java : ");
 Student st = new Student("Dharmit", 19);
 st.display();

 // Learning static and final variable difference
 System.out.println("\nLearning static and final variable difference : ");
 new Counter();
 new Counter();
 new Counter();

 System.out.println("Counter : " + Counter.count);

 // Learning Instance Variable
 System.out.println("\nLearning Instance Variable : ");
 Main e = new Main();
 e.id = 9;
 e.name = "Dharmit";
 System.out.println("Name : " + e.name + "," + "ID : " + e.id);

 // Calling static function in java
 System.out.println("\nCalling static function in java : ");
 System.out.println("Sum : " + Calculator.add(8, 15));

 // Inner and Outer Classes
 System.out.println("\nInner and Outer classes : ");
 Outer.Inner obj = new Outer().new Inner();
 obj.display();
 }
}
```

Outcome

```
PS C:\Users\DHARMIT SHAH\Desktop\Ass2> java Main.java
Defining class in java :
name : Dharmit, age : 19

Learning static and final variable difference :
Object : 1
Object : 2
Object : 3
Counter : 3

Learning Instance Variable :
Name : Dharmit,ID : 9

Calling static function in java :
Sum : 23

Inner and Outer classes :
Inner Class Method
```

Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

```java
class Student {
    String name;
    int age;
    int id;

    Student () {
        name = "Name";
        age = 0;
        id = 0;
    }

    Student (String n, int a, int i) {
        name = n;
        age = a;
        id = i;
    }

    void display () {
        System.out.println(x: "Student Details : ");
        System.out.println("Name : " + name);
        System.out.println("Age  : " + age);
        System.out.println("ID   : " + id);
    }
}

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        Student st = new Student(n: "Dharmit", a: 19, i: 157);
        st.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\Ass2\Exercise> java Ex1.java
Student Details :
Name : Dharmit
Age  : 19
ID   : 157
```

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

2. Develop a **Mobile Phone Inventory System** using different constructors to initialize mobile details and create duplicate object records.

```
class Mobile {
    String brand;
    String model;
    String CPU;
    int RAM;
    int storage;
    int price;

    Mobile () {
        brand = "Brand name";
        model = "Model";
        CPU = "CPU name";
        RAM = 0;
        storage = 0;
        price = 0;
    }

    Mobile (String b, String m, String c, int r, int s, int p) {
        brand = b;
        model = m;
        CPU = c;
        RAM = r;
        storage = s;
        price = p;
    }

    Mobile (Mobile m) {
        this.brand = m.brand;
        this.model = m.model;
        this.CPU = m.CPU;
        this.RAM = m.RAM;
        this.storage = m.storage;
        this.price = m.price;
    }

    void display () {
        System.out.println("Mobile Details : ");
        System.out.println("Brand   : " + brand);
        System.out.println("CPU     : " + CPU);
        System.out.println("RAM     : " + RAM + " GB");
        System.out.println("Storage : " + storage + " GB");
        System.out.println("Price   : Rs." + price);
        System.out.println();
    }
}
```

```java
public class Ex2 {
 public static void main (String[] args) {
 Mobile m1 = new Mobile("Realme", "Realme GT 6", "Qualcomm Snapdragon 8s Gen3", 12, 256, 35000);
 Mobile m2 = new Mobile(m1);
 m1.display();
 m2.display();
 }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\Ass2\Exercise> java Ex2.java
Mobile Details :
Brand   : Realme
CPU     : Qualcomm Snapdragon 8s Gen3
RAM     : 12 GB
Storage : 256 GB
Price   : Rs.35000

Mobile Details :
Brand   : Realme
CPU     : Qualcomm Snapdragon 8s Gen3
RAM     : 12 GB
Storage : 256 GB
Price   : Rs.35000
```